AGNEY K RAJEEV

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About Me

I am a **5th-year year Integrated MSc. Student** at **NISER Bhubaneswar**. I'm majoring in **Physics** (and minoring in **Computer Science**) with a particular inquisitiveness for **Statistical Mechanics, Soft Condensed Matter Physics and Biological Physics**. My specific curiosity is in the **physics of living matter** which lies at the **intersection of Physics, Biology and Computational Science**. I am interested in understanding complex biological relations and systems such as bird flocks, bacterial colonies, proteins in media etc., creating interaction-based physics models for the same and using computer simulations (aided by machine learning protocols if necessary) to evolve, observe and analyse emergent behaviours.

Research Experience

- **Computational Studies of The Ising Model** — *Jun 2023 - Aug 2023*
  Summer Internship — NISER Bhubaneswar
  - Guide: Dr. A. V. Anil Kumar
  - Studied the 2D Ising model using the Metropolis Algorithm
  - Determined the existence of a phase transition and evaluated the Critical Temperature
  - Attempted to incorporate non-reciprocal interactions and random fields to the model
- **Predicting Protein Oligomericity from Amino Acid Sequence** — *Jan 2024 - Sep 2024*
  Semester Project — NISER Bhubaneswar
  - Guide: Dr. Subhankar Mishra
  - Attempted to create a machine learning model for predicting the oligomerization states of proteins from amino acid sequences
  - Achieved near state-of-the-art results in classifying fluorescent proteins into monomers and oligomers
- **Predicting the Stress Tensor for a simulated molecular system** — *May 2024 - Jul 2024*
  Summer Project — IISER Kolkata
  - Guide: Dr. Neelanjana Sengupta
  - Seeked to develop an efficient alternative to predicting Stress Tensor for biomolecules in a solution using deep learning
  - Successfully obtained novel preliminary results for the atomic stress tensor of a water box using node regression learning in Graph Neural Networks
- **Effects of non-reciprocal interactions in synchronization of oscillating ecological metapopulations** — *Jun 2025 - ongoing*
  MSc. Thesis — NISER Bhubaneswar
  - Guide: Dr. A. V. Anil Kumar
  - Integrated non-reciprocal dispersal to a metapopulation model in Ising universality
  - Attempt to observe any change in phase transitions, critical values or universality class
  - Also study the dynamic behaviour of the new model

Education

- **National Institute of Science Education and Research** — *2021 - ongoing*
  Integrated MSc. — Bhubaneswar, India
  - CGPA: 8.95/10.0
- **Chinmaya Vidyalaya Kolazhy** — *2009 - 2021*
  Higher Secondary School — Thrissur, India
  - Class XII: 97.0/100.0
  - Class X: 95.0/100.0

Publications

C=Conference, P=Preprint

[P.1] Agney K. Rajeev, & A. V. Anil Kumar. (2024). Ising model with non-reciprocal interactions.

[C.1] Rajeev, A.K., B, J.J.K. &; Mishra, S.. (2025). Predicting Oligomeric states of Fluorescent Proteins using Mamba. *Proceedings of the 6th Northern Lights Deep Learning Conference (NLDL)*, in *Proceedings of Machine Learning Research* 265:204-212.

Skills

- **Programming Languages:** Python, Julia, Fortran, Javascript
- **Data Science & Machine Learning:** Pandas, PyTorch
- **Mathematical & Statistical Tools:** MATLAB, Mathematica
- **Other Tools & Technologies:** Arduino, SeeLab

Honors and Awards

- **Rank 528** *2025*
  Graduate Aptitude Test in Engineering(GATE)
  - National-level entrance exam in India for postgraduate admission to Master's and PhD programs in engineering, technology, science, architecture, and humanities.
- **Semester IV Topper** *2023*
  NISER Bhubaneswar
  - GPA: 9.36
- **Scholar** *2021*
  Innovation in Science Pursuit for Inspired Research (INSPIRE)
  - Innovative programme sponsored and managed by the Department of Science & Technology for attraction of talent to Science
  - Eligible for top 1% students in Class XII exams

Relevant Coursework

- **Physics :** Classical Mechanics I & II, Mathematical Methods I & II, Quantum MechanicsI & II, Statistical Mechanics, Introduction to Condensed Matter Physics, NonEquilibrium Statistical Mechanics, NonLinear Dynamics and Chaos, Phase Transitions and Critical Phenomena
- **Computer Science :** Introductory Python Programming I & II, Theory of Computation, Discrete Structures and Computation, Design and Analysis of Algorithms, Machine Learning, Complexity Theory, Parametrized algorithms
- **Biology :** Introductory BioPhysics

Additional Information

Languages: Malayalam(Mother Tongue), English(Fluent), Hindi(Fluent), Spanish(Very Basic)
Interests: Table Tennis, Piano, Coding, Cryptography